

(d) Electromagnetic induction	
Students should:	
6.15	know that a voltage is induced in a conductor or a coil when it moves through a magnetic field or when a magnetic field changes through it and describe the factors that affect the size of the induced voltage
6.16	describe the generation of electricity by the rotation of a magnet within a coil of wire and of a coil of wire within a magnetic field, and describe the factors that affect the size of the induced voltage
6.17P describe the structure of a transformer and understand that a transformer changes the size of an alternating voltage by having different numbers of turns on the input and output sides	
6.18P explain the use of step-up and step-down transformers in the large-scale generation and transmission of electrical energy	
6.19P know and use the relationship between input (primary) and output (secondary) voltages and the turns ratio for a transformer: $\frac{\text{input (primary) voltage}}{\text{output (secondary) voltage}} = \frac{\text{primary turns}}{\text{secondary turns}}$	
6.20P know and use the relationship: input power = output power $V_p I_p = V_s I_s$ for 100% efficiency	

7 Radioactivity and particles

The following sub-topics are covered in this section.

- (a) Units
 - (b) Radioactivity
 - (c) Fission and fusion
-